

Quantum-Limited Measurement Engineering

A Practical Handbook for Noise, Readout, Control, and Metrology

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June 2026

This handbook develops the vocabulary, mathematics, and experimental workflow needed to design and assess measurements near quantum noise limits.

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Chapter 1

Foundations for Quantum-Limited Measurement

1.1 What a measurement does

Every measurement seeks an unknown physical parameter θ , such as displacement, force, phase, frequency, magnetic field, temperature, photon number, or qubit state. A sensor couples this parameter to a probe and produces a physical response that electronics or photodetectors convert into a classical record $y(t)$, where t denotes time. An estimator $\hat{\theta}[y]$ then maps the complete record to an inferred value of the parameter.

The complete process can be written as

$$\theta \longrightarrow \rho_\theta \longrightarrow \text{transduced field or pointer} \longrightarrow \text{record } y(t) \longrightarrow \hat{\theta}[y]. \quad (1.1)$$

The state ρ_θ describes the probe after the parameter has been encoded. A classical model can represent this state with a probability distribution over possible configurations, whereas the quantum description uses a density operator containing the probabilities and coherences needed to predict measurement outcomes. The final expression $\hat{\theta}[y]$ makes the dependence of the reported estimate on the complete record and the chosen analysis rule explicit.

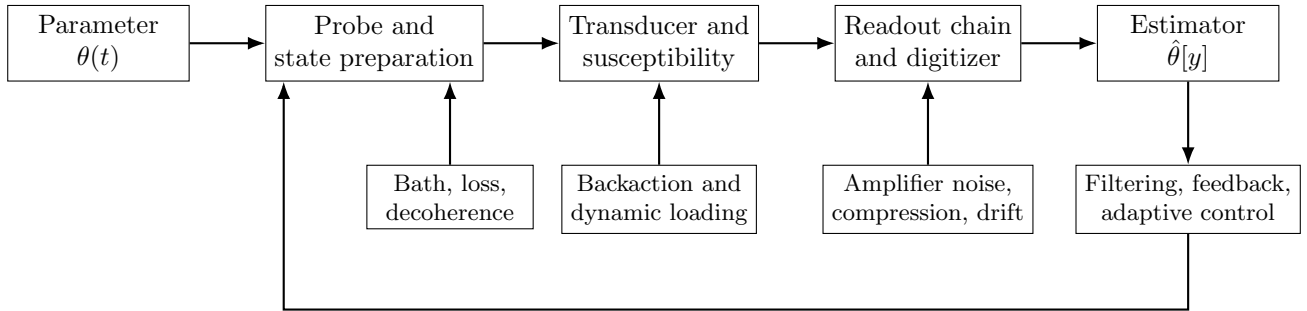


Figure 1.1: The measurement chain.

The chain separates physical encoding from information transport and inference. This separation matters because noise at the digitizer differs from noise at the first amplifier input, at the transducer output, or at the original signal input. A valid noise budget converts every contribution to one declared reference plane and one declared physical variable.

1.2 Probability, averages, and uncertainty

Repeated measurements generally produce different outcomes. A random variable Y describes those possible outcomes, and $p(y | \theta)$ gives the probability density for observing y when the parameter has value θ . The vertical

bar means “conditioned on.” The expectation value, also called the mean, is

$$\langle Y \rangle = \int y p(y | \theta) dy. \quad (1.2)$$

The variance measures the spread around the mean,

$$\text{Var}(Y) = \langle (Y - \langle Y \rangle)^2 \rangle. \quad (1.3)$$

Taking the square root gives the standard deviation. For a fluctuating quantity A , the root mean square (RMS) value is $\sqrt{\langle A^2 \rangle}$, which equals the standard deviation when the fluctuation has zero mean.

Two quantities can fluctuate together. Their covariance is

$$\text{Cov}(A, B) = \langle (A - \langle A \rangle)(B - \langle B \rangle) \rangle. \quad (1.4)$$

A positive covariance means that the quantities tend to increase together, and a negative covariance means that one tends to decrease as the other increases. Quantum measurement design often uses a controlled negative covariance to cancel part of the total noise.

An estimator has bias

$$\text{Bias}(\hat{\theta}) = \langle \hat{\theta} \rangle - \theta. \quad (1.5)$$

Its mean square error (MSE) combines spread and bias,

$$\text{MSE}(\hat{\theta}) = \langle (\hat{\theta} - \theta)^2 \rangle = \text{Var}(\hat{\theta}) + \text{Bias}(\hat{\theta})^2. \quad (1.6)$$

1.3 Accuracy, precision, sensitivity, and resolution

Together, accuracy, precision, sensitivity, and resolution describe complementary properties of a measurement. Accuracy quantifies agreement with a traceable or accepted value after calibration, whereas precision describes the spread of repeated estimates. Sensitivity measures how strongly the detector output changes with the input parameter, and resolution identifies the smallest change that a declared procedure can distinguish under specified conditions. A precise instrument can remain inaccurate when a calibration offset is present, and a sensitive transducer can remain noisy when its output chain adds large fluctuations.

Term	Operational question	Typical mathematical quantity
Accuracy	How close is the estimate to a traceable or accepted value?	Calibration error, bias, systematic uncertainty
Precision	How tightly do repeated estimates cluster?	Variance or standard deviation
Sensitivity	How strongly does the output respond to the input?	Transfer slope $\partial \langle y \rangle / \partial \theta$ or transfer function $H_{y\theta}$
Resolution	What minimum change can the full procedure distinguish?	Detection threshold, confidence interval, discrimination error
Uncertainty	What range of values remains compatible with the data and calibration model?	Standard uncertainty, confidence interval, credible interval

Calibration connects the instrument output to the input parameter. A calibration uncertainty propagates into every input referred sensitivity. Drift describes a slow change in gain, offset, phase, resonance frequency, or another operating parameter. Stationarity means that the statistical properties remain constant over the analysis interval. Spectral methods require stationarity or an explicit treatment of time variation.

1.4 Frequency, bandwidth, and linear response

A time dependent signal can be described by its frequency components. Ordinary frequency f is measured in hertz, and angular frequency is $\omega = 2\pi f$ in radians per second. The bandwidth is the range of frequencies admitted by the detector, filter, or estimator. Increasing bandwidth allows faster changes to be observed and usually admits more noise.

A linear system maps an input θ to an output y according to

$$y[\omega] = H_{y\theta}[\omega]\theta[\omega] + n_y[\omega]. \quad (1.7)$$

The complex transfer function $H_{y\theta}$ contains gain and phase, and the additive term n_y represents output noise. A susceptibility describes the response of a physical coordinate to its conjugate drive; for example, the mechanical susceptibility χ_{xx} maps force to displacement.

Input referring converts an output noise into the equivalent fluctuation of the original parameter. This operation allows voltage noise, displacement noise, force noise, and digitizer noise to be compared on a shared basis. The reference plane specifies the location in the measurement chain at which gain, loss, and noise are defined.

1.5 Minimal quantum vocabulary

A quantum state is represented by a density operator ρ . It is Hermitian, has nonnegative eigenvalues, and satisfies $\text{Tr}(\rho) = 1$. A pure state can be written as $\rho = |\psi\rangle\langle\psi|$. Mixed states represent classical uncertainty over quantum states or entanglement with unobserved degrees of freedom.

An observable is represented by a Hermitian operator A . Its expectation value is

$$\langle A \rangle = \text{Tr}(\rho A), \quad (1.8)$$

and its variance is

$$\text{Var}(A) = \langle A^2 \rangle - \langle A \rangle^2. \quad (1.9)$$

The commutator

$$[A, B] = AB - BA \quad (1.10)$$

measures the incompatibility of two operations or observables. A Hamiltonian H is the energy operator that generates time evolution. Closed system evolution is represented by a unitary operator U , which preserves state normalization and inner products. The Robertson uncertainty relation states

$$\Delta A \Delta B \geq \frac{1}{2} |\langle [A, B] \rangle|. \quad (1.11)$$

This relation constrains the state variances of incompatible observables. Measurement backaction is a dynamical disturbance created by the coupling between the detector and the measured system. The two ideas are related, although they describe different parts of the measurement problem.

Many optical, microwave, mechanical, acoustic, and magnonic excitations can be modeled as bosonic modes. Their annihilation and creation operators a and $a^\dagger$ satisfy

$$[a, a^\dagger] = 1. \quad (1.12)$$

The number operator $a^\dagger a$ counts excitations, whereas the vacuum state contains zero excitations yet retains quantum fluctuations. Two real field components called quadratures are

$$X = \frac{a + a^\dagger}{\sqrt{2}}, \quad P = \frac{a - a^\dagger}{i\sqrt{2}}, \quad [X, P] = i. \quad (1.13)$$

Vacuum has $\text{Var}(X) = \text{Var}(P) = 1/2$ in this convention. A squeezed state reduces one quadrature variance below $1/2$ and increases the conjugate variance so that the uncertainty relation remains satisfied.

Decoherence is the loss of phase relationships that carry quantum information. Dissipation transfers energy to an environment, and dephasing randomizes phase with little or no energy exchange. A quantum channel describes the complete state transformation caused by interaction, loss, noise, measurement, or control.

1.6 Noise and limit vocabulary

The measurement channel carries the desired response together with fluctuations that weaken the selected inference. Thermal noise reflects finite temperature and environmental occupation, technical noise arises from classical imperfections such as vibration, source phase noise, drift, leakage, or electronics, and quantum noise enters through the probe state, detector interaction, and channel used to extract information.

After the noise terms have been referred to a common variable and reference plane, the signal to noise ratio (SNR) compares the chosen signal metric with the corresponding noise metric. A power spectral density (PSD) expresses variance per unit frequency, and its square root gives the amplitude spectral density (ASD) in units such as metres per square root hertz or volts per square root hertz. Power ratios R are expressed in decibels as $10 \log_{10} R$. For equal impedances, an amplitude ratio A uses $20 \log_{10} A$. Chapter 2 establishes the spectral conventions used throughout this handbook.

Quantum benchmarks describe different constraints on the measurement chain. The standard quantum limit (SQL) is architecture dependent, so its equation changes with the measurement task and the assumptions imposed on the probe, detector, and estimator. Independent counting or independent probes define the shot noise limit (SNL). Quantum Fisher information (QFI) quantifies the maximum local information available about a parameter in a quantum state, and the quantum Cramér–Rao bound (QCRB) converts that information into a lower bound on estimator variance under stated regularity conditions. A quantum nondemolition (QND) measurement preserves the selected observable under repeated interrogation in the ideal model.

The central distinction

A quantum limit becomes meaningful after the parameter, observable, estimator, resource count, bandwidth, spectral convention, loss model, and allowed controls have been fixed. Different limits answer different questions, so their equations cannot be substituted for one another without checking their assumptions.

1.7 The measurement contract

A measurement contract records the task before hardware optimization begins. It identifies the parameter and units, operating point, transfer function, useful bandwidth or interrogation time, estimator, error metric, counted resources, allowed controls, and selected benchmark.

Measurement contract template

The measurement estimates θ over the band $\mathcal{B}$ with the estimator $\hat{\theta}[y]$. Every noise contribution is referred to the input variable θ , and the resulting uncertainty is compared with benchmark B under matched bandwidth, averaging time, probe resources, duty cycle, loss, and estimator assumptions.

As a comprehensive ledger, the contract accounts for photons, atoms, energy, total experimental time, repetitions, ancillas, preparation time, dead time, discarded trials, and feedback resources whenever they affect the comparison. It also specifies whether squeezing, entanglement, adaptive measurements, auxiliary modes, and postselection are admissible.

1.8 A taxonomy of quantum limits

Quantum limits fall into several classes, and each class constrains a different stage of the measurement chain.

Class	Representative benchmark	Engineering meaning
Fundamental estimation bound	Quantum Cramér–Rao, Holevo Cramér–Rao, Helstrom, and quantum Chernoff bounds	Optimizes over allowed measurements for a specified state family, prior structure, and loss function.

Architecture dependent standard quantum limit	Shot noise scaling, projection noise scaling, and imprecision plus backaction balance	Applies after the probe state, detector topology, measured observable, or correlation structure has been restricted. A changed architecture can surpass the benchmark.
Detector channel limit	One half quantum of added noise for a high gain phase preserving amplifier	Constrains the map between input and output modes. A phase sensitive channel changes the task by selecting one quadrature.
Dynamical or thermodynamic bound	Quantum speed limit and Landauer erasure cost	Constrains evolution time or irreversible heat under declared physical assumptions.
Implementation floor	Thermal occupation, insertion loss, pump noise, drift, and finite efficiency	Describes the realized apparatus. These terms often dominate before a more fundamental bound becomes observable.

A statement such as “three decibels below the standard quantum limit” remains incomplete until the relevant benchmark is named. Source squeezing below vacuum, detector imprecision below a displacement benchmark, and total force sensitivity below a conventional continuous measurement limit represent distinct achievements.

1.9 Three connected performance layers

The probe layer determines how strongly the parameter is encoded. Its metrics include quadrature variance, spin squeezing, entanglement depth, purity, coherence, and quantum Fisher information. The transport layer determines how much encoded information reaches the detector. Its metrics include escape efficiency, insertion loss, mode matching, thermal occupation, and amplifier added noise. The inference layer determines how much of the arriving information is extracted. Its metrics include estimator bias, variance, prior sensitivity, calibration uncertainty, and latency.

Performance at the quantum limit requires all three layers to preserve the advantage. Strong source squeezing can disappear after passive loss. A low noise amplifier can feed an inefficient estimator. An optimal estimator cannot recover information that escaped into an unobserved environment.

1.10 Scope of device-specific extensions

This handbook concentrates on quantum measurement physics, readout chains, estimation, control, and the engineering evidence needed to support a limit specification. Device-specific treatments of microwave dielectric loss, quasiparticle dynamics, electromagnetic participation, nonlinear resonator design, fabrication process variation, bosonic code hardware, and full-wave parameter extraction remain implementation extensions. Their measured consequences enter the present framework as loss, thermal occupation, susceptibility, nonlinearity, leakage, state-transition, and channel-efficiency parameters. This scope preserves a common measurement architecture across optical, microwave, mechanical, atomic, acoustic, and magnonic platforms without replacing the specialized device physics required by any one platform.

Chapter 2

Noise Spectra and Input Referred Budgets

2.1 Time and frequency descriptions

A time trace shows when a fluctuation occurs, whereas a spectrum shows how its variance is distributed across frequency. This handbook uses the Fourier transform

$$A[\omega] = \int_{-\infty}^{\infty} A(t) e^{i\omega t} dt. \quad (2.1)$$

For a stationary quantity, define the fluctuation $\delta A = A - \langle A \rangle$. The unsymmetrized power spectral density is

$$S_{AB}[\omega] = \int_{-\infty}^{\infty} \langle \delta A(t) \delta B(0) \rangle e^{i\omega t} dt. \quad (2.2)$$

Operator order matters for quantum quantities, so $S_{AB}[\omega]$ and $S_{BA}[-\omega]$ can differ. The symmetrized spectrum is

$$\bar{S}_{AB}[\omega] = \frac{1}{2} (S_{AB}[\omega] + S_{BA}[-\omega]). \quad (2.3)$$

Its diagonal form gives the variance through

$$\langle \delta A^2 \rangle = \int_{-\infty}^{\infty} \frac{d\omega}{2\pi} \bar{S}_{AA}[\omega]. \quad (2.4)$$

Equations (2.2) through (2.4) use a two sided spectrum in angular frequency. For an even stationary spectrum, the corresponding one sided engineering spectrum in hertz is

$$S_{AA}^{(1)}(f) = 2\bar{S}_{AA}[2\pi f], \quad f > 0. \quad (2.5)$$

White noise with constant $S_{AA}^{(1)}$ across a bandwidth Δf has RMS fluctuation

$$\Delta A_{\text{rms}} = \sqrt{S_{AA}^{(1)} \Delta f}. \quad (2.6)$$

The associated amplitude spectral density is $\sqrt{S_{AA}^{(1)}}$.

Spectral labels

A complete spectral comparison identifies whether the spectrum is one sided or two sided, whether its horizontal axis uses hertz or radians per second, and whether a quantum spectrum is symmetrized or unsymmetrized. These labels prevent factor of two and factor of 2π errors.

2.2 Vacuum fluctuations and thermal occupation

The quadratures in Equation (1.13) have vacuum variance $1/2$. A thermal mode at frequency f and temperature T has mean occupation

$$\bar{n}_{\text{th}} = \frac{1}{\exp(hf/k_{\text{B}}T) - 1}, \quad (2.7)$$

where h is Planck's constant and k_{B} is Boltzmann's constant. The symmetrized noise number is

$$N_{\text{sym}} = \bar{n}_{\text{th}} + \frac{1}{2}. \quad (2.8)$$

The dimensionless ratio $hf/(k_{\text{B}}T)$ reveals whether thermal occupation is large or small. Low frequency mechanical modes can remain highly occupied at millikelvin temperatures, whereas gigahertz microwave modes can approach their ground state in the same cryostat.

A resistance R produces the quantum Nyquist voltage spectrum

$$\bar{S}_{VV}[\omega] = R\hbar|\omega| \coth\left(\frac{\hbar|\omega|}{2k_{\text{B}}T}\right). \quad (2.9)$$

Its one sided low frequency limit is $4k_{\text{B}}TR$, which is the classical Johnson noise result. This connection between dissipation and equilibrium fluctuations is an example of the fluctuation dissipation theorem [1, 2].

2.3 Input referred noise

The linear model in Equation (1.7) gives an equivalent input noise spectrum

$$S_{\theta\theta}^{\text{eq}}[\omega] = \frac{S_{yy}^n[\omega]}{|H_{y\theta}[\omega]|^2}. \quad (2.10)$$

Each physical contribution requires its own transfer function. After conversion, force noise, displacement noise, voltage noise, and digitizer noise can be expressed in the common units of θ .

Independent noise terms add as

$$S_{\theta\theta}^{\text{tot}} = \sum_j S_{\theta\theta,j}^{\text{eq}}. \quad (2.11)$$

Correlated terms require cross spectra,

$$S_{\theta\theta}^{\text{tot}} = \sum_j S_{jj}^{\text{eq}} + 2 \sum_{j < k} \text{Re } S_{jk}^{\text{eq}}. \quad (2.12)$$

The cross term can increase or decrease total noise. Omitting it can hide a useful quantum correlation or create an apparent cancellation that lacks a physical basis.

2.4 Passive loss and information loss

A passive channel with power transmission η mixes the signal mode with an environmental mode. The output quadrature variance is

$$V_{\text{out}} = \eta V_{\text{in}} + (1 - \eta) \left(\bar{n}_{\text{b}} + \frac{1}{2} \right), \quad (2.13)$$

where $\bar{n}_{\text{b}}$ is the environmental occupation. For a vacuum environment and an input squeezing parameter r ,

$$\frac{V_{\text{out}}}{V_{\text{vac}}} = \eta e^{-2r} + 1 - \eta. \quad (2.14)$$

Finite transmission therefore sets an observed squeezing floor even when the source produces a much smaller intrinsic variance.

The same channel contributes input referred symmetrized added noise

$$N_{\text{loss}} = \frac{1-\eta}{\eta} \left(\bar{n}_{\text{b}} + \frac{1}{2} \right). \quad (2.15)$$

Loss before the first informative gain stage carries a larger penalty than equal loss after large gain, since early loss removes encoded information before amplification.

2.5 Matched filtering and task level performance

A sensitivity curve gains practical meaning only after it is combined with the signal waveform. For a known waveform $s(t)$ in stationary Gaussian noise with one sided spectrum $S_n^{(1)}(f)$, the optimum linear matched filter gives

$$\text{SNR}^2 = 4 \int_0^\infty \frac{|\tilde{s}(f)|^2}{S_n^{(1)}(f)} df. \quad (2.16)$$

A narrow noise reduction outside the signal band contributes little to this integral. Consequently, a complete report pairs the spectral curve with an integrated signal to noise ratio, estimator variance, or detection error appropriate to the task.

Chapter 3

Continuous Linear Measurement

3.1 Detector model and mechanical response

A continuous detector produces an uninterrupted record of a system observable. Linear operation means that small changes in the observable produce proportional changes in the detector output. A general interaction between a measured coordinate x and a detector force F is

$$H_{\text{int}} = -xF. \quad (3.1)$$

The Hamiltonian H_{int} determines how the detector both acquires information and perturbs the system. Its output I responds according to

$$I[\omega] = I_0[\omega] + \chi_{IF}[\omega]x[\omega], \quad (3.2)$$

where I_0 is the output fluctuation in the absence of the signal and χ_{IF} is the detector gain. Referring I_0 back to the coordinate gives the imprecision

$$x_{\text{imp}}[\omega] = \frac{I_0[\omega]}{\chi_{IF}[\omega]}. \quad (3.3)$$

Imprecision describes uncertainty in the inferred coordinate. Backaction describes the motion caused by fluctuations of the detector force F .

The measured system responds to signal, thermal, and backaction forces through

$$x[\omega] = \chi_{xx}[\omega] (F_{\text{sig}}[\omega] + F_{\text{th}}[\omega] + F_{\text{ba}}[\omega]). \quad (3.4)$$

For a damped harmonic oscillator with mass m , resonance frequency ω_m , and energy damping rate Γ_m ,

$$\chi_{xx}[\omega] = \frac{1}{m(\omega_m^2 - \omega^2 - i\Gamma_m\omega)}. \quad (3.5)$$

Setting $\omega_m = 0$ and $\Gamma_m = 0$ gives the ideal free mass susceptibility $\chi_{xx} = -1/(m\omega^2)$.

3.2 Total added displacement noise

The detector contributes three terms to the measured displacement spectrum,

$$\bar{S}_{xx}^{\text{add}} = \bar{S}_{xx}^{\text{imp}} + |\chi_{xx}|^2 \bar{S}_{FF}^{\text{ba}} + 2 \text{Re} [\chi_{xx}^* \bar{S}_{xF}]. \quad (3.6)$$

Together, these terms combine detector imprecision with motion driven by backaction and the cross spectrum that correlates them. Depending on its sign and phase, the cross spectrum can raise or lower the total noise.

The complete measurement also contains thermal and technical terms,

$$\bar{S}_{xx}^{\text{meas}} = \bar{S}_{xx}^{\text{th}} + \bar{S}_{xx}^{\text{add}} + \bar{S}_{xx}^{\text{tech}}. \quad (3.7)$$

The distinction between added noise and total measured noise is essential. Even a quantum-limited detector can operate in an experiment whose thermal motion remains far above the target sensitivity.

In the canonical normalization used here, imprecision, backaction, and their correlation satisfy

$$\bar{S}_{xx}^{\text{imp}} \bar{S}_{FF}^{\text{ba}} - |\bar{S}_{xF}|^2 \geq \frac{\hbar^2}{4}. \quad (3.8)$$

The expression assumes negligible reverse gain and an effective detector normalization. More general linear response inequalities include the response functions explicitly [2].

3.3 The conventional continuous measurement standard quantum limit

The conventional continuous measurement standard quantum limit follows after two restrictions are imposed. First, the imprecision and backaction correlation is set to zero. Second, the detector saturates the uncertainty product in Equation (3.8). Minimization of Equation (3.6) over measurement strength then gives

$$\bar{S}_{xx}^{\text{imp,opt}} = \frac{\hbar |\chi_{xx}|}{2}, \quad |\chi_{xx}|^2 \bar{S}_{FF}^{\text{ba,opt}} = \frac{\hbar |\chi_{xx}|}{2}. \quad (3.9)$$

The two contributions are equal at the optimum, and their sum becomes

$$\boxed{\bar{S}_{xx}^{\text{SQL}}[\omega] = \hbar |\chi_{xx}[\omega]|}. \quad (3.10)$$

Increasing probe power generally lowers imprecision and raises backaction. Decreasing power reverses that trade. The minimum therefore appears at the crossing of the two contributions when the ideal assumptions apply.

For a free mass,

$$\bar{S}_{xx}^{\text{SQL}}[\omega] = \frac{\hbar}{m\omega^2}, \quad S_{xx}^{(1),\text{SQL}}(f) = \frac{2\hbar}{m(2\pi f)^2}. \quad (3.11)$$

These equations use a two sided symmetrized angular frequency spectrum and a one sided hertz spectrum, respectively. Time domain expressions depend on the sampling and estimator convention, so the spectral form usually provides the clearest engineering reference [3, 4].

3.4 Force referred bounds and the dissipative floor

Dividing the displacement limit by $|\chi_{xx}|^2$ gives the uncorrelated force standard quantum limit,

$$\bar{S}_{FF}^{\text{SQL}} = \frac{\hbar}{|\chi_{xx}|}. \quad (3.12)$$

Useful correlations can lower this quantity. For a broad class of linear force measurements, the correlation optimized ultimate quantum limit (UQL) takes the form

$$\boxed{\bar{S}_{FF}^{\text{UQL}}[\omega] = \hbar |\text{Im } \chi_{xx}^{-1}[\omega]|}. \quad (3.13)$$

The imaginary part of the inverse susceptibility represents dissipation. Applied to the oscillator in Equation (3.5),

$$\bar{S}_{FF}^{\text{UQL}} = \hbar m \Gamma_m |\omega|. \quad (3.14)$$

Dissipation therefore sets a residual force noise scale after ideal correlation engineering. The ideal free mass has zero dissipation in this model, so practical loss, control bandwidth, and stability become the relevant limitations [5, 6].

3.5 Thermal decoherence and quantum cooperativity

A mode with thermal occupation $\bar{n}_{\text{th}}$ and damping rate Γ_m exchanges excitations with its environment at a rate of order $\Gamma_m \bar{n}_{\text{th}}$ when the occupation is large. For a cavity enhanced measurement with linearized coupling rate G , cavity linewidth κ , and mechanical linewidth Γ_m , the cooperativity is

$$C = \frac{4G^2}{\kappa\Gamma_m}. \quad (3.15)$$

The high temperature quantum cooperativity is commonly written as

$$C_q = \frac{C}{\bar{n}_{\text{th}}}. \quad (3.16)$$

The condition $C_q \gtrsim 1$ indicates that measurement induced quantum dynamics compete with thermal decoherence. Definitions can differ by factors near unity when zero point terms, detection efficiency, or sideband asymmetry are included [7].

Red detuned sideband cooling increases the damping of a mechanical mode through its coupling to a colder electromagnetic bath. An approximate final occupation is

$$\bar{n}_f \simeq \frac{\Gamma_m \bar{n}_{\text{th}} + \Gamma_{\text{opt}} \bar{n}_{\text{min}}}{\Gamma_m + \Gamma_{\text{opt}}}, \quad \bar{n}_{\text{min}} \simeq \left(\frac{\kappa}{4\omega_m} \right)^2. \quad (3.17)$$

The optical damping rate is Γ_{opt} . Pump absorption, frequency noise, and nonequilibrium environments often set a higher measured floor than the ideal sideband limit [8, 9, 7].

Chapter 4

Engineering Workflow for Reaching a Conventional Standard Quantum Limit

4.1 Step 1. Benchmark specification

The benchmark equation is established before hardware modifications begin. A continuous position meter may use Equation (3.10), the force referred form in Equation (3.12), or a system specific interferometer expression. An ensemble sensor is benchmarked against a declared independent probe resource count, and an amplifier comparison distinguishes phase preserving from phase sensitive operation.

A useful comparison ratio is

$$\mathcal{R}_{\text{SQL}}(\omega) = \frac{S_{\text{meas}}(\omega)}{S_{\text{SQL}}(\omega)}. \quad (4.1)$$

4.2 Step 2. Transduction gain and susceptibility calibration

The transfer function $H_{y\theta}$ is calibrated with a traceable injected signal, and the system susceptibility is quantified independently whenever the apparatus permits. Dynamic backaction, optical spring effects, saturation, and feedback can alter the susceptibility during operation; dividing by a fixed model can therefore introduce a hidden systematic error into the inferred sensitivity.

The calibration record contains complex gain, phase, uncertainty, drift rate, operating power, detuning, temperature, and feedback state. Repeating the calibration after noise acquisition tests whether the measurement remained stationary.

4.3 Step 3. Input referred noise budget

The input referred budget is organized by physical origin,

$$S_{\theta\theta}^{\text{tot}} = S_{\text{thermal}} + S_{\text{imprecision}} + S_{\text{backaction}} + S_{\text{correlation}} + S_{\text{technical}} + S_{\text{digitizer}}. \quad (4.2)$$

Each term is supported by either a measured transfer function or a model constrained by independent data. Cross spectra enter the budget whenever two channels share a source or an intentionally generated quantum correlation. Any model subtraction propagates the full calibration covariance into the residual uncertainty.

4.4 Step 4. Thermal occupation control

Bath temperature, mode frequency, material loss, thermal anchoring, line filtering, radiation shielding, optical absorption, and active cooling provide the main controls over thermal occupation. Cryostat temperature alone gives incomplete evidence, since the mode can decouple from the cold stage or absorb pump power. The relevant quantity is the mode occupation inferred from sideband asymmetry, calibrated thermometry, equipartition in a validated classical regime, or a quantum calibrated noise source.

Microwave attenuator placement balances field thermalization against cable loss and first stage gain. Mechanical systems require joint consideration of intrinsic linewidth, thermal force, response time, and active cooling bandwidth.

4.5 Step 5. Loss before the first informative gain

Equation (2.15) quantifies the penalties introduced by cables, circulators, filters, couplers, windows, mode mismatch, and detector inefficiency. Preinteraction loss degrades state preparation, interaction loss removes information after encoding, and postinteraction loss reduces collection and detection efficiency.

The path between the transducer and first gain stage often has the highest leverage. Improving that path can exceed the benefit obtained from stronger source squeezing or additional downstream gain.

4.6 Step 6. Quantum-limited first stage gain

For high gain, a quantum-limited phase preserving first stage approaches one half input referred added quantum. The cascade relation is

$$N_{\text{add,sys}} = N_1 + \frac{N_2}{G_1} + \frac{N_3}{G_1 G_2} + \dots, \quad (4.3)$$

where G_j is the power gain of stage j and N_j is its input referred added noise. Sufficient first stage gain suppresses later electronics noise, although excessive gain can reduce dynamic range.

Characterization covers bandwidth, compression power, pump leakage, reverse noise, gain ripple, intermodulation, magnetic sensitivity, and long term stability. Noise calibration is repeated across the operating powers used in the experiment [10, 11, 12, 13].

4.7 Step 7. Measurement strength sweep

A measurement strength sweep provides a direct diagnostic of the imprecision and backaction trade. An ideal position detector often follows

$$S_{xx}^{\text{imp}} \propto P^{-1}, \quad S_{FF}^{\text{ba}} \propto P, \quad (4.4)$$

where P is probe power. Under the ideal scaling, the total added noise passes through a minimum. A monotonic curve usually indicates that one component remains below the calibration floor or that a technical contribution changes with power.

The sweep spans enough dynamic range to identify both trends without entering a regime dominated by heating, instability, or nonlinear response. Fits include the uncertainty of the power calibration and the possibility that the susceptibility changes with power.

4.8 Step 8. Estimator selection

Estimator choice follows the signal and noise model. Matched filtering applies to a known waveform, Wiener filtering to stationary linear estimation, Kalman filtering to state space models, Bayesian inference to informative priors, and smoothing to analyses that admit records acquired after the estimation time. Reported performance includes estimator bias and prior dependence together with variance.

Feedback can change susceptibility, dynamic range, and stability. An external sensitivity claim is based on an out of loop record or an equivalent open loop reconstruction, since in loop noise squashing can make the recorded signal appear quieter without increasing the information about an external force [14, 15, 16].

4.9 Step 9. Physical attribution of noise terms

Temperature sweeps identify thermal contributions, probe power sweeps separate imprecision from backaction, and detuning sweeps reveal cavity filtering and dynamic backaction. Pump phase sweeps test quadrature mixing, deliberate signal injection validates the transfer function, and cross correlation between independent detectors reveals common technical noise.

Attribution to quantum noise gains credibility when absolute magnitude, parameter scaling, and cross spectrum agree with one physical model across several independent sweeps.

4.10 Step 10. Uncertainty and margin

The logarithmic margin is defined by

$$M_{\text{dB}}(\omega) = 10 \log_{10} \left[\frac{S_{\text{SQL}}(\omega)}{S_{\text{meas}}(\omega)} \right]. \quad (4.5)$$

Positive values indicate a measured power spectral density below the selected standard quantum limit. The confidence interval includes calibration gain, mass, temperature, linewidth, detection efficiency, spectral estimation bias, and finite averaging. A positive central value smaller than the systematic uncertainty remains consistent with the limit and provides insufficient evidence for a definitive sub-SQL claim.

Recommended evidence
Robust evidence includes a calibrated total noise spectrum, a component budget, a measurement strength sweep, an independent susceptibility measurement, an uncertainty band, and a task level quantity such as integrated signal to noise ratio or estimator variance. Raw spectra acquired before model subtraction remain part of the record.

Chapter 5

Methods That Change the Conventional Standard Quantum Limit Assumptions

5.1 Squeezed probes

A single mode squeezed state has quadrature variances

$$V_- = \frac{1}{2}e^{-2r}, \quad V_+ = \frac{1}{2}e^{2r}, \quad V_- V_+ = \frac{1}{4}, \quad (5.1)$$

where r is the squeezing parameter. The lower variance V_- improves a signal encoded in the aligned quadrature, and the higher variance V_+ preserves the uncertainty product. Useful performance depends on squeeze factor, squeeze angle, source escape efficiency, propagation efficiency, detector efficiency, and phase lock bandwidth.

A phase error $\delta\phi$ mixes the two quadratures,

$$V_{\text{meas}} = V_- \cos^2 \delta\phi + V_+ \sin^2 \delta\phi. \quad (5.2)$$

Strong antisqueezing therefore increases sensitivity to phase noise. Metrological credit is based on the detected variance after every loss and phase error.

5.2 Phase sensitive amplification

A phase preserving amplifier treats both quadratures equally and can be represented by

$$b = \sqrt{G}a + \sqrt{G-1}d^\dagger, \quad (5.3)$$

where d is an auxiliary mode required to preserve the output commutator. At high gain, the auxiliary contribution produces at least one half input referred quantum of added noise [10].

A phase sensitive amplifier selects one quadrature,

$$X_{\text{out}} = \sqrt{G}X_{\text{in}}, \quad P_{\text{out}} = \frac{1}{\sqrt{G}}P_{\text{in}}. \quad (5.4)$$

The amplified quadrature can approach zero added noise in the ideal model. This advantage requires prior knowledge or tracking of the signal phase, and phase drift mixes the deamplified quadrature into the result.

5.3 Variational readout and engineered correlations

Homodyne detection measures a selected field quadrature by interfering the signal with a strong phase reference called a local oscillator. A frequency dependent homodyne angle can select an output combination that carries useful

imprecision and backaction correlations. The optimization becomes

$$\min_{\zeta(\omega), P, \Delta} \left[\bar{S}_{xx}^{\text{imp}} + |\chi|^2 \bar{S}_{FF}^{\text{ba}} + 2 \operatorname{Re} (\chi^* \bar{S}_{xF}) \right], \quad (5.5)$$

where ζ is the readout angle and Δ controls detuning or another correlation parameter. Optical loss and phase error weaken the cancellation. Experimental validation includes the cross spectrum or an equivalent quadrature covariance model [17, 6].

5.4 Frequency dependent squeezing

The optimum squeeze angle changes across frequency when imprecision dominates one band and radiation pressure backaction dominates another. A filter cavity or an equivalent transformation rotates the squeezing ellipse so that its low noise axis follows the detector optimum.

Filter cavity detuning, linewidth, round trip loss, mode matching, squeeze angle control, and interferometer operating point determine the broadband gain. The Laser Interferometer Gravitational Wave Observatory (LIGO) has demonstrated broadband quantum enhancement with frequency dependent squeezing, illustrating the value of optimizing the full signal band as one system [18].

5.5 Backaction evading quadrature measurement

For a mechanical mode with annihilation operator b , a rotating quadrature is

$$X_\phi = x_{\text{zpf}} (b e^{-i\phi} + b^\dagger e^{i\phi}), \quad (5.6)$$

where x_{zpf} is the zero point displacement scale. A balanced two tone interaction can measure one rotating quadrature and direct backaction into its conjugate. This backaction evading (BAE) architecture can surpass the conventional standard quantum limit for the selected quadrature.

Counterrotating terms, pump imbalance, detuning error, finite sideband resolution, and parametric instability leak backaction into the measured channel. Verification requires quadrature resolved spectra and a measurement strength sweep that separates imprecision from residual backaction [19, 20].

5.6 Quantum nondemolition observables and speed meters

An observable O supports repeated quantum nondemolition measurement when the interaction and free evolution preserve it in the ideal continuous model,

$$[H_{\text{int}}, O] = 0, \quad [H_{\text{sys}}, O] = 0. \quad (5.7)$$

Photon number, a collective spin component, and a rotating oscillator quadrature are common examples. Stroboscopic protocols can preserve the observable at selected times even when continuous preservation is unavailable.

A speed meter samples a momentum like or velocity like quantity, often by interrogating position twice with opposite backaction response. Low frequency radiation pressure contributions then cancel. Optical loss, delay mismatch, and finite storage time set the residual disturbance.

5.7 Coherent quantum noise cancellation

Coherent quantum noise cancellation (CQNC) uses an auxiliary system with a response opposite to the primary sensor. If

$$\chi_a[\omega] \simeq -\chi_m[\omega] \quad (5.8)$$

and the couplings match, the residual backaction response scales as

$$S_{\text{ba, res}} \propto |\chi_m + \chi_a|^2 S_{FF}^{\text{ba}}. \quad (5.9)$$

Negative effective mass spin ensembles and engineered optical modes can provide the auxiliary response. Resonance frequency, damping, coupling, phase, and bath occupation must match across the target band. Auxiliary decoherence then fixes the cancellation floor [21, 22].

5.8 Optimal estimation, feedback, and smoothing

A continuously monitored quantum state can be represented by a stochastic master equation, which updates the state according to both deterministic evolution and the random measurement record. Filtering uses data available up to the present. Smoothing or retrodiction combines earlier and later data to estimate an intermediate state. Feedback applies a control action based on the filtered estimate [14, 15].

These methods improve conditional purity, state stabilization, dynamic range, and task level estimation. Their performance remains bounded by information lost to unobserved channels. A feedback cooled oscillator can show reduced in loop motion, yet a force sensitivity analysis must include the feedback modified susceptibility and the actuator noise.

5.9 Reservoir engineering

Reservoir engineering modifies the environment so that dissipation prepares or protects a useful state. Controlled dissipation can cool a mode, stabilize squeezing, or protect a subspace. The main controls are dissipative coupling, pump detuning, bath purity, auxiliary linewidth, and spectral separation.

This approach changes state preparation and dynamics before inference. A fair comparison therefore includes pump resources, preparation time, residual heating, and every new loss channel introduced by the engineered environment.

Chapter 6

Quantum Estimation and Metrological Bounds

6.1 Classical Fisher information

A probability model $p(y | \theta)$ changes as the parameter θ changes. The score function $\partial_\theta \ln p(y | \theta)$ measures the local sensitivity of the log likelihood to that change. Its mean square defines the classical Fisher information,

$$F_C(\theta) = \int p(y | \theta) [\partial_\theta \ln p(y | \theta)]^2 dy. \quad (6.1)$$

For an unbiased estimator based on ν independent repetitions, the classical Cramér–Rao bound is

$$\text{Var}(\hat{\theta}) \geq \frac{1}{\nu F_C(\theta)}. \quad (6.2)$$

The bound is local because it depends on derivatives evaluated near the true parameter. Attainability generally improves with many repetitions and a regular statistical model.

6.2 Quantum Fisher information and the quantum Cramér–Rao bound

A quantum state can be measured in many bases, and each measurement produces a different classical probability model. Quantum Fisher information is the maximum Fisher information available from the state over all permitted measurements. The quantum Cramér–Rao bound is

$$\boxed{\text{Var}(\hat{\theta}) \geq \frac{1}{\nu F_Q(\theta)}}. \quad (6.3)$$

For a pure state generated by

$$U_\theta = \exp(-i\theta t G), \quad (6.4)$$

where G is the generator of parameter encoding, the quantum Fisher information is

$$F_Q = 4t^2 \text{Var}(G). \quad (6.5)$$

State design can therefore be interpreted as the design of generator variance. The measurement must still extract that information, and the estimator must operate in the regime where the local bound is meaningful [23, 24].

6.3 Shot noise scaling and Heisenberg scaling

Consider N probes used in each of ν independent repetitions. Separable probes with a fixed single-probe generator range have quantum Fisher information that grows linearly with probe number, whereas an ideal collectively entangled

probe can support quadratic growth,

$$F_Q^{\text{sep}} \propto N, \quad F_Q^{\text{ent}} \propto N^2. \quad (6.6)$$

Substitution into the quantum Cramér–Rao bound gives the shot noise limit scaling

$$\Delta\theta_{\text{SNL}} \propto \frac{1}{\sqrt{\nu N}}, \quad (6.7)$$

and the canonical Heisenberg limit (HL) scaling

$$\boxed{\Delta\theta_{\text{HL}} \propto \frac{1}{\sqrt{\nu N}}}. \quad (6.8)$$

The factor $\nu^{-1/2}$ comes from independent repetition. The factor N^{-1} comes from the ideal collective generator scaling. Loss and Markovian noise often restore the asymptotic $N^{-1/2}$ dependence, although a useful constant-factor gain can remain [24, 25, 26].

Resource accounting determines whether an apparent gain survives at the system level. The count includes interrogation time, state preparation, dead time, discarded trials, ancillas, adaptive rounds, and total energy whenever these resources differ between strategies.

6.4 Local bounds, priors, and global ambiguity

The quantum Cramér–Rao bound describes local curvature near one parameter value. A periodic phase can produce several likelihood peaks, and a weak signal can enter a threshold regime where local curvature predicts an unrealistically small error. Prior information also changes the task by assigning different probabilities to different parameter values.

The quantum Ziv–Zakai bound (QZZB) relates global mean square error to the difficulty of discriminating parameter values separated by a finite interval. Its structure can be written as

$$\Sigma \geq \frac{1}{2} \int_0^\infty \tau \mathcal{V}\{g(\tau)\} d\tau, \quad (6.9)$$

where Σ is the mean square error, $\mathcal{V}$ denotes valley filling, and $g(\tau)$ combines the prior overlap with the minimum discrimination error between states separated by τ [27, 28]. A robust analysis compares a local information bound with a global Bayesian or discrimination based bound whenever the likelihood is broad or multimodal.

6.5 Multiparameter estimation

A sensor may need to estimate several quantities at once, such as field amplitude and phase, temperature and loss, or a spatial field and its gradient. Let $\boldsymbol{\theta}$ collect the parameters. The symmetric logarithmic derivative (SLD) operators L_i satisfy

$$\partial_i \rho_\theta = \frac{L_i \rho_\theta + \rho_\theta L_i}{2}. \quad (6.10)$$

They define the quantum Fisher information matrix (QFIM),

$$[\mathbf{F}_Q]_{ij} = \text{Tr} \left[\rho_\theta \frac{L_i L_j + L_j L_i}{2} \right]. \quad (6.11)$$

The formal matrix bound is

$$\text{Cov}(\hat{\boldsymbol{\theta}}) \succeq \frac{1}{\nu} \mathbf{F}_Q^{-1}. \quad (6.12)$$

Different parameters can require incompatible optimal measurements, which prevents simultaneous saturation of every matrix element. For the general asymptotic problem, the Holevo Cramér–Rao bound (HCRB) gives the standard weighted scalar benchmark [29, 30, 31]. Engineering design therefore begins with the desired scalar objective. Estimating a global average, a gradient, and every local value are separate tasks even when they use the same sensor network.

6.6 State discrimination and detection

Some readout problems ask which of several hypotheses generated the record. For two quantum states ρ_0 and ρ_1 with prior probabilities p_0 and p_1 , the minimum single shot error is the Helstrom bound,

$$P_{e,\min} = \frac{1}{2} (1 - \|p_0\rho_0 - p_1\rho_1\|_1), \quad (6.13)$$

where $\|\cdot\|_1$ is the trace norm. The optimum measurement is a positive operator valued measure, which generalizes projective measurement and assigns a positive operator to each possible outcome.

Repeated independent copies produce an asymptotic error exponent governed by the quantum Chernoff bound (QCB),

$$\xi_{\text{QCB}} = -\ln \left[\min_{0 \leq s \leq 1} \text{Tr}(\rho_0^s \rho_1^{1-s}) \right]. \quad (6.14)$$

Qubit state assignment, target detection, and quantum illumination often fit discrimination bounds more naturally than a variance based standard quantum limit [32, 33, 34].

6.7 Information, dynamics, and thermodynamics

Quantum measurement engineering also encounters limits that constrain information transfer, evolution time, or heat. The Holevo accessible information bound limits the classical information obtainable from a quantum ensemble,

$$I_{\text{acc}} \leq \chi = S(\rho) - \sum_x p_x S(\rho_x), \quad (6.15)$$

where $S(\rho) = -\text{Tr}(\rho \ln \rho)$ is the von Neumann entropy [35, 34].

For isolated evolution toward an orthogonal state, the Mandelstam–Tamm and Margolus–Levitin quantum speed limits give

$$\tau \geq \max \left[\frac{\pi \hbar}{2\Delta E}, \frac{\pi \hbar}{2\langle E \rangle - E_0} \right]. \quad (6.16)$$

Here ΔE is the energy standard deviation and E_0 is the ground state energy. Open dynamics and constrained controls require modified bounds [36, 37, 38].

Landauer’s principle gives the minimum heat associated with quasistatic erasure of one symmetric classical bit into a bath at temperature T ,

$$Q_{\text{diss}} \geq k_B T \ln 2. \quad (6.17)$$

This thermodynamic result becomes relevant when control, reset, memory, and feedback resources are included in a complete measurement system [39].

Chapter 7

Nonclassical Probes and Coherence Engineering

7.1 Collective spin and projection noise

An ensemble of N effective spin one half particles can be described by a collective spin vector $\mathbf{J}$. A coherent spin state points along a mean direction and has equal quantum fluctuations in the two transverse directions. Ramsey interferometry converts a small phase into a rotation of this collective vector, so transverse projection noise sets the independent probe sensitivity.

The Wineland metrological squeezing parameter is

$$\xi_R^2 = \frac{N(\Delta J_\perp)^2}{|\langle \mathbf{J} \rangle|^2}. \quad (7.1)$$

A value $\xi_R^2 < 1$ certifies sensitivity below coherent spin state projection noise for the associated Ramsey protocol. The ideal single interrogation phase uncertainty is

$$\Delta\phi = \frac{\xi_R}{\sqrt{N}}. \quad (7.2)$$

The mean spin length in the denominator charges contrast loss and decoherence against the noise reduction. A reduction of $\Delta J_\perp$ alone gives incomplete evidence for metrological gain [40, 41, 26].

7.2 Entangled probes and information certification

Multipartite entanglement can increase generator variance and quantum Fisher information. For collective linear generators, $F_Q > N$ certifies useful particle entanglement under standard assumptions. The available detector and estimator determine how much of that information appears in the reported sensitivity.

A Greenberger–Horne–Zeilinger (GHZ) state provides the ideal N^2 quantum Fisher information scaling and high fragility to loss. Spin squeezed states provide a smaller theoretical gain with greater resilience. The preferred probe follows from a joint optimization of preparation fidelity, interrogation time, decoherence, readout resolution, and duty cycle.

7.3 Dynamical decoupling and filter functions

A controlled two level sensor accumulates phase

$$\phi = \gamma \int_0^T y(t)b(t) dt, \quad (7.3)$$

where $b(t)$ is the signal, γ is the coupling coefficient, and $y(t) = \pm 1$ is the modulation created by control pulses. The frequency response is

$$Y(\omega) = \int_0^T y(t) e^{i\omega t} dt. \quad (7.4)$$

Pulse timing places sensitivity near selected frequencies and suppresses other noise bands. This technique is especially useful for alternating current (AC) signals. It can reduce sensitivity to direct current (DC) or very low frequency signals, and its practical performance depends on finite pulse duration, control error, heating, and pulse timing jitter [42].

Dynamical decoupling commonly increases the useful interrogation time and improves the sensitivity prefactor. Scaling changes require additional resources such as entanglement, adaptive control, or error correction.

7.4 Distributed quantum sensing

A sensor network can estimate local parameters θ or a global function $q = \mathbf{w}^\top \theta$. Entanglement across nodes provides its clearest advantage when the desired observable is global. Independent local parameters often favor separable states and local measurements, whereas a weighted average can benefit from a shared squeezed or entangled mode [43, 44, 45].

Channel transmission, phase synchronization, clock distribution, mode matching, node dependent coupling, local readout efficiency, and estimator weights determine the useful gain. Network loss acts on the probe state before quantum Fisher information is evaluated or enters the analysis explicitly as a quantum channel.

7.5 Quantum error correction assisted metrology

Markovian noise is memoryless on the time scale of the model and can be described by a Lindblad master equation. The operators L_k identify the error channels. For a signal Hamiltonian $H(\omega) = \omega G$, define the Lindblad operator span

$$\mathcal{S} = \text{span}_{\mathbb{C}} \left\{ I, L_k, L_k^\dagger, L_k^\dagger L_j \right\}_{j,k}. \quad (7.5)$$

Ideal error corrected metrology can recover Heisenberg scaling when the generator lies outside this span,

$$\boxed{G \notin \mathcal{S}}. \quad (7.6)$$

This condition is often called the Hamiltonian outside the Lindblad span criterion. It assumes sufficiently clean ancillas and fast accurate control [46, 47, 48].

An engineering budget must include syndrome cycle time, ancilla initialization, gate errors, leakage, decoder latency, and retained signal response. A code that removes the generator also removes the desired phase, so noise rejection and signal preservation must be verified together.

7.6 Adaptive measurement

Adaptive protocols update the preparation, interrogation time, or measurement basis using earlier outcomes. They can resolve phase wrapping, maintain operation near maximum slope, and approach a local bound across a broad prior. Latency, model error, digitization, and actuator resolution enter the procedure as physical constraints.

A fair comparison counts every adaptive round, probe, reset, and discarded trial. Posterior width, interval coverage, and estimator bias accompany any quoted sensitivity.

Chapter 8

Quantum-Limited Readout Chains

8.1 Amplifier quantum limits

A phase preserving amplifier treats both field quadratures equally and, in the high gain limit, obeys

$$N_{\text{add}} \geq \frac{1}{2}, \quad T_N \geq \frac{hf}{2k_B}. \quad (8.1)$$

The quantity N_{add} is input referred added noise in quanta, and T_N is the corresponding high gain noise temperature under the convention used here. Input vacuum or thermal fluctuations also contribute to the complete system noise. Alternative noise temperature definitions can shift constant factors without changing the half quantum added noise theorem [10, 2, 13].

Phase sensitive amplification can approach zero added noise in one selected quadrature. Frequency conversion, nonreciprocity, and parametric gain can be combined, yet every passive element before gain mixes in vacuum or thermal fluctuations.

8.2 Cascade budgeting and preamplifier loss

For a chain of gains G_j and added noises N_j , Equation (4.3) gives the total input referred contribution. A passive path of transmission η followed by an amplifier with noise N_A gives device referred added noise

$$N_{\text{add,dev}} = \frac{1-\eta}{\eta} \left(\bar{n}_b + \frac{1}{2} \right) + \frac{N_A}{\eta}. \quad (8.2)$$

Large first stage gain suppresses noise from a later high electron mobility transistor (HEMT) amplifier or room temperature electronics. It cannot restore information removed by preamplifier loss.

Under the quadrature convention of Chapter 2, an effective single quadrature detection efficiency is

$$\eta_{\text{det}} = \frac{1}{1 + 2N_{\text{add}}}. \quad (8.3)$$

A zero added noise phase sensitive chain gives $\eta_{\text{det}} = 1$. An ideal phase preserving chain with $N_{\text{add}} = 1/2$ gives $\eta_{\text{det}} = 1/2$ for this information metric.

8.3 Qubit readout efficiency

A qubit readout maps two quantum states onto two distributions of an integrated detector record. For approximately Gaussian hypotheses, define

$$\text{SNR}^2(t) = \frac{[\mu_1(t) - \mu_0(t)]^2}{\sigma_0^2(t) + \sigma_1^2(t)}, \quad (8.4)$$

where μ_j and σ_j^2 are the conditional mean and variance for state j . The operational measurement efficiency is

$$\eta_m = \frac{1}{4\Gamma_\varphi} \frac{d}{dt} \text{SNR}^2(t), \quad \eta_m \leq 1, \quad (8.5)$$

where Γ_φ is the measurement induced dephasing rate produced by the same readout field. Unit efficiency means that every dephasing channel carrying qubit information is monitored. Different signal to noise ratio conventions change the numerical factor, so each reported efficiency includes its defining equation [2, 49, 50].

Resonator linewidth, dispersive shift, drive envelope, integration filter, collection efficiency, amplifier mode, gain, saturation, isolation, and state relaxation control the result. Stronger drive can increase state separation and also increase leakage, nonlinear response, dressed dephasing, and transitions outside the computational subspace.

8.4 Calibration methods

Microwave calibration methods include hot and cold loads, calibrated noise sources, shot noise tunnel junctions, sideband asymmetry, qubit dephasing, and coherent tone injection. Optical methods include balanced homodyne vacuum calibration, detector quantum efficiency measurements, calibrated attenuation, and shot noise scaling.

Each calibration report identifies the reference plane, gain, bandwidth, impedance, occupation convention, and uncertainty. Compression and drift tests use the same pump, signal, and bias conditions employed during the measurement.

8.5 Readout chain control table

Control	Primary benefit	Secondary cost	Diagnostic
First stage gain	Suppresses later amplifier noise	Compression, gain ripple, pump leakage	Noise versus gain and input power
Preamplifier insertion loss	Preserves encoded information	Packaging and isolation complexity	Cold transmission calibration and mode occupation
Phase sensitive mode	Removes the half quantum penalty in one quadrature	Phase tracking and reduced signal generality	Quadrature noise ellipse and phase sweep
Readout power	Raises state separation or measurement rate	Backaction, transitions, heating, and nonlinearity	Signal to noise ratio and dephasing versus power
Integration filter	Maximizes extracted information	Model dependence and latency	Cross validation on independent records
Isolation and directionality	Protects the device from reverse noise	Added loss and bandwidth limits	Reverse noise and dephasing measurement

Chapter 9

Worked Engineering Examples

9.1 Example 1. Thermal occupation across platforms

Consider a 5 MHz mechanical mode at 10 mK. Equation (2.7) gives

$$\frac{hf}{k_B T} = 0.023996, \quad \bar{n}_{\text{th}} \simeq 41.18. \quad (9.1)$$

The mode remains far above its ground state despite the millikelvin bath. Passive cooling, a higher resonance frequency, or active sideband cooling is required.

A 5 GHz microwave mode at 20 mK gives

$$\frac{hf}{k_B T} = 11.998, \quad \bar{n}_{\text{th}} \simeq 6.16 \times 10^{-6}. \quad (9.2)$$

Its equilibrium occupation is negligible. Residual photons then point toward imperfect filtering, weak thermalization, infrared loading, or pump leakage.

9.2 Example 2. One decibel of loss before an ideal amplifier

Take a 6 GHz readout at 20 mK. One decibel of passive loss has

$$\eta = 10^{-1/10} = 0.7943, \quad \bar{n}_b \simeq 5.59 \times 10^{-7}. \quad (9.3)$$

Place an ideal phase preserving amplifier after the loss, so $N_A = 1/2$. Equation (8.2) gives

$$N_{\text{loss}} = \frac{1 - \eta}{\eta} \left(\bar{n}_b + \frac{1}{2} \right) \simeq 0.1295, \quad (9.4)$$

$$\frac{N_A}{\eta} \simeq 0.6295, \quad (9.5)$$

$$N_{\text{add,dev}} \simeq 0.7589. \quad (9.6)$$

The equivalent single quadrature efficiency is

$$\eta_{\text{det}} \simeq \frac{1}{1 + 2(0.7589)} = 0.397. \quad (9.7)$$

One passive decibel lowers the ideal phase preserving efficiency from one half to about 0.397. Reducing preamplifier loss can therefore provide more value than adding downstream gain.

9.3 Example 3. Ramsey projection noise and spin squeezing

Consider $N = 10^6$ independent atoms and $\nu = 100$ repetitions. The shot noise phase scale is

$$\Delta\phi_{\text{SNL}} = \frac{1}{\sqrt{\nu N}} = 10^{-4} \text{ rad.} \quad (9.8)$$

Six decibels of metrological spin squeezing corresponds to

$$\xi_R^2 = 10^{-6/10} = 0.2512, \quad \xi_R = 0.5012. \quad (9.9)$$

If contrast and duty cycle remain unchanged,

$$\Delta\phi = \frac{\xi_R}{\sqrt{\nu N}} \simeq 5.01 \times 10^{-5} \text{ rad.} \quad (9.10)$$

A real clock comparison replaces the ideal assumptions with observed contrast, local oscillator noise, state preparation time, dead time, and readout fidelity. Recent optical clock measurements illustrate the importance of this complete accounting [51].

9.4 Example 4. Symbolic optimum measurement strength

Suppose imprecision and backaction follow

$$\bar{S}_{xx}^{\text{imp}} = \frac{A}{P}, \quad \bar{S}_{FF}^{\text{ba}} = BP, \quad (9.11)$$

with zero correlation. The total added displacement noise is

$$\bar{S}_{xx}^{\text{add}}(P) = \frac{A}{P} + |\chi|^2 BP. \quad (9.12)$$

Differentiate with respect to power,

$$\frac{d\bar{S}_{xx}^{\text{add}}}{dP} = -\frac{A}{P^2} + |\chi|^2 B. \quad (9.13)$$

Setting this derivative to zero gives

$$P_{\text{opt}} = \sqrt{\frac{A}{|\chi|^2 B}}. \quad (9.14)$$

Substitution yields

$$\bar{S}_{xx}^{\text{add},\text{min}} = 2|\chi|\sqrt{AB}. \quad (9.15)$$

A quantum ideal uncorrelated detector has $AB = \hbar^2/4$, which recovers $\hbar|\chi|$. Experimental fits to A and B expose excess detector noise and reveal whether the crossing occurs at the expected power.

9.5 Example 5. Detected squeezing after loss and phase error

Assume a source produces 10 dB of quadrature squeezing and 15 dB of antisqueezing. The vacuum normalized source variances are

$$R_- = 10^{-10/10} = 0.100, \quad R_+ = 10^{15/10} = 31.62. \quad (9.16)$$

Let the total transmission be $\eta = 0.80$ and the phase error be $\delta\phi = 1^\circ$. Loss first gives

$$R'_- = \eta R_- + 1 - \eta = 0.280, \quad R'_+ = \eta R_+ + 1 - \eta = 25.50. \quad (9.17)$$

Equation (5.2) then gives

$$R_{\text{meas}} = R'_- \cos^2 \delta\phi + R'_+ \sin^2 \delta\phi \simeq 0.2877. \quad (9.18)$$

The detected squeezing is

$$-10 \log_{10}(R_{\text{meas}}) \simeq 5.41 \text{ dB.} \quad (9.19)$$

The source value of 10 dB becomes about 5.4 dB at the detector. In this example, loss produces the larger penalty, and antisqueezing makes the remaining phase error measurable.

Chapter 10

Validation of Quantum Limit Claims

10.1 Levels of claim

Claim	Minimum evidence
Quantum noise observed	Absolute magnitude and parameter scaling agree with a quantum model, and independent sweeps constrain thermal and technical alternatives.
Quantum-limited component	A named component, such as amplifier added noise or detector imprecision, reaches its bound at a declared reference plane.
Standard quantum limit level detector	Total added noise reaches the specified standard quantum limit after imprecision, backaction, and correlations are included.
Sub standard quantum limit measurement	Calibrated total task relevant noise or estimator variance falls below the specified standard quantum limit with a positive uncertainty margin.
Quantum enhanced metrology	Achieved precision exceeds the best stated classical or separable strategy under equal resources, loss, duty cycle, and estimator assumptions.
Fundamental bound approached	The realized state, measurement, and estimator nearly saturate the relevant quantum Cramér–Rao, Holevo, Helstrom, or other task level bound.

A low noise source, a quantum-limited amplifier, and a sub-SQL total estimator are separate results.

10.2 Common false positives

One component can lie below a standard quantum limit even when the total remains above it. In loop feedback can suppress the recorded fluctuations without improving sensitivity to an external signal. Model subtraction can create a low residual when calibration covariance is omitted. Source squeezing can exceed detected squeezing. Postselection can improve conditional precision and reduce the success rate. A narrow spectral improvement can disappear after integration over the signal waveform.

Each risk has a corresponding control. Raw and processed spectra are presented together, and feedback claims are supported by an out of loop monitor or equivalent reconstruction. Calibration covariance is propagated through model subtraction. Success probability, dead time, and discarded trials accompany postselected results. The strongest spectral point is evaluated alongside an integrated signal to noise ratio, estimator variance, or detection error.

10.3 Claim checklist

Check	Required statement
C1	The measured parameter, reference plane, units, operating point, and signal band are explicit.
C2	The benchmark equation and every defining assumption are explicit.
C3	One sided or two sided form, hertz or radians per second, and quantum operator ordering are explicit.
C4	Probe number, energy, averaging time, repetition count, duty cycle, ancillas, preparation time, and discarded trials are counted.
C5	Thermal occupation and environmental temperature are measured or bounded at the relevant mode.
C6	Passive loss and detection efficiency are measured at the operating frequency and power.
C7	Amplifier added noise, gain, bandwidth, and compression are calibrated at the declared reference plane.
C8	Imprecision, backaction, and cross correlation terms are included whenever the benchmark requires them.
C9	Technical noise sources have independently measured transfer functions or upper bounds.
C10	The estimator, prior, bias, interval definition, and uncertainty propagation are stated.
C11	Feedback results include the modified susceptibility and an out of loop record or equivalent reconstruction.
C12	Squeezing results include detected variance, antisqueezing, phase noise, and total efficiency.
C13	Entanglement results include the witness or quantum Fisher information criterion and the measurement that extracts the gain.
C14	Subtraction, postselection, and discarded data are disclosed with uncertainty or success probability.
C15	The standard quantum limit ratio or metrological gain includes a systematic uncertainty band.

Decision rule

A quantum limit conclusion is robust when it survives a change of analysis basis. The calibrated spectrum, time domain estimator, and parameter sweep tied to the physical noise model then support the same result.

Appendix A

Master Table of Quantum Limits and Engineering Relations

#	Benchmark or relation	Equation	Class	Assumptions, interpretation, and engineering response	Reference
1	Poisson shot noise current	$\Delta I_{\text{rms}} = \sqrt{2qI\Delta f}$, $S_I^{(1)} = 2qI$	Independent counting benchmark	Assumes independent charge arrivals. The Fano factor $F = \text{Var}(N)/\langle N \rangle$ compares count variance with mean count. A value below unity gives $S_I^{(1)} = 2qIF$ and indicates sub Poisson statistics.	Standard
2	Coherent beam phase shot noise	$S_{\phi\phi}^{\text{coh}} = 1/(4\dot{N}) = \hbar\omega/(4\bar{P})$	Optical standard quantum limit	Assumes a coherent state, a declared quadrature convention, and ideal phase readout. Squeezing or engineered correlations can lower the signal quadrature variance.	[52, 2]
3	Phase preserving amplifier limit	$N_{\text{add}} \geq 1/2$, $T_N \geq \hbar f/(2k_B)$	Detector channel limit	Applies to high gain linear amplification of both quadratures. Phase sensitive operation selects one quadrature and can approach zero added noise in that quadrature.	[10, 2]
4	Free mass displacement standard quantum limit	$\bar{S}_{xx}^{\text{SQL}} = \hbar/(m\omega^2)$	Continuous measurement limit	Assumes uncorrelated quantum ideal imprecision and backaction. Correlations, speed meters, quantum nondemolition variables, or backaction evasion change the architecture.	[3, 4]
5	Canonical imprecision and backaction product	$\bar{S}_{xx}^{\text{imp}} \bar{S}_{FF}^{\text{ba}} \geq \hbar^2/4$	Detector relation	Uses an effective detector normalization with zero useful correlation. General response inequalities include detector gain and reverse gain.	[2, 53]
6	Canonical Heisenberg phase scaling	$\Delta\phi_{\text{HL}} \sim \frac{1}{\sqrt{\nu N}}$	Metrology scaling	Assumes the conventional N probe generator normalization, ideal unitary encoding, and consistent resource accounting. Decoherence can return the scaling to the shot noise form.	[24, 25]
7	Separable probe shot noise scaling	$\Delta\phi_{\text{SNL}} \sim 1/\sqrt{\nu N}$	Metrology standard quantum limit	Applies to independent probes with a fixed single probe generator range. Squeezing or entanglement can improve the prefactor or scaling.	[24, 26]
8	Quantum Cramér–Rao bound	$\text{Var}(\hat{\theta}) \geq 1/(\nu F_Q)$	Fundamental estimation bound	Describes local unbiased estimation with regularity and attainability conditions. State preparation, interaction, measurement, and estimator must be considered together.	[23]
9	Correlated detector noise constraint	$\bar{S}_{xx}^{\text{imp}} \bar{S}_{FF}^{\text{ba}} - \bar{S}_{xF} ^2 \geq \hbar^2/4$	Detector relation	Canonical form for a linear detector. A controlled cross spectrum can reduce total noise over selected frequencies.	[2]
10	Operational qubit quantum nondemolition readout efficiency	$\eta_m = [\text{dSNR}^2/\text{d}t]/(4\Gamma_\varphi) \leq 1$	Readout efficiency	Uses $\text{SNR}^2 = (\mu_1 - \mu_0)^2/(\sigma_0^2 + \sigma_1^2)$. Unit efficiency requires collection of every dephasing channel that carries qubit information.	[2, 49]
11	Landauer erasure cost	$Q_{\text{diss}} \geq k_B T \ln 2$	Thermodynamic bound	Applies to quasistatic erasure of one symmetric bit into a bath at temperature T . General memories require a free energy statement.	[39]
12	Holevo accessible information bound	$I_{\text{acc}} \leq \chi = S(\rho) - \sum_x p_x S(\rho_x)$	Information bound	Applies to a fixed quantum ensemble and a classical message. It limits information accessible by any measurement on the ensemble.	[35, 34]
13	Helstrom binary error bound	$P_{e,\text{min}} = \frac{1}{2}[1 - \ p_0\rho_0 - p_1\rho_1\ _1]$	Hypothesis testing bound	Assumes fixed binary states and priors. The optimum positive operator valued measurement attains the bound.	[32]
14	Frequency domain displacement standard quantum limit	$\bar{S}_{xx}^{\text{SQL}} = \hbar \chi_{xx} $	Continuous measurement limit	Assumes linear response, uncorrelated imprecision and backaction, and a quantum ideal detector.	[2, 3]
15	Quantum speed limit	$\tau \geq \max[\pi\hbar/(2\Delta E), \pi\hbar/(2(\langle E \rangle - E_0))]$	Dynamical bound	Applies to pure state orthogonalization under a time independent isolated Hamiltonian. Open systems and constrained controls require modified forms.	[36, 37, 38]

#	Benchmark or relation	Equation	Class	Assumptions, interpretation, and engineering response	Reference
16	Quantum Chernoff error exponent	$\xi_{\text{QCB}} = -\ln \min_{0 \leq s \leq 1} \text{Tr}(\rho_0^s \rho_1^{1-s})$	Hypothesis testing bound	Gives the asymptotic binary discrimination exponent for independent copies.	[33]
17	Quantum Ziv–Zakai bound	$\Sigma \geq \frac{1}{2} \int_0^\infty \tau \mathcal{V}\{g(\tau)\} d\tau$	Bayesian estimation bound	Uses an explicit prior and finite parameter separations. It remains useful at low signal to noise ratio and with multimodal likelihoods.	[27, 28]
18	Multiparameter symmetric logarithmic derivative quantum Cramér–Rao bound	$\text{Cov}(\hat{\theta}) \succeq \mathbf{F}_Q^{-1}/\nu$	Multiparameter estimation	Simultaneous saturation can fail when the optimal measurements for different parameters are incompatible.	[29]
19	Holevo Cramér–Rao bound	$\text{Tr}[W \text{Cov}] \geq C_H(W)/\nu$	Multiparameter estimation	Supplies a tight weighted asymptotic benchmark under broad regularity conditions when measurement incompatibility matters.	[31, 30]
20	Thermal occupation	$\bar{n}_{\text{th}} = [\exp(hf/k_B T) - 1]^{-1}$	Implementation floor	The mode temperature can differ from the cryostat temperature. Improve thermalization, filtering, mode frequency, or active cooling.	[7]
21	Quantum Nyquist spectrum	$\bar{S}_{VV} = R\hbar \omega \coth(\hbar \omega /2k_B T)$	Fluctuation dissipation relation	Uses a two sided symmetrized angular frequency convention. The low frequency one sided form becomes $4k_B TR$.	[1]
22	Passive loss mixing	$V_{\text{out}} = \eta V_{\text{in}} + (1 - \eta)(\bar{n}_b + 1/2)$	Quantum channel relation	Uses a beam splitter model. Low preamplifier loss and low environmental occupation preserve encoded information.	[2]
23	Cascaded amplifier noise	$N_{\text{sys}} = N_1 + N_2/G_1 + N_3/(G_1 G_2) + \dots$	Engineering relation	Uses power gains and input referred noise quanta. First stage gain and noise usually dominate the chain.	[13]
24	Force referred standard quantum limit	$\bar{S}_{FF}^{\text{SQL}} = \hbar/ \chi_{xx} $	Continuous measurement limit	Gives the equivalent force noise associated with the uncorrelated displacement standard quantum limit.	[2]
25	Dissipative ultimate quantum limit	$\bar{S}_{FF}^{\text{UQL}} = \hbar \text{Im} \chi_{xx}^{-1} $	Linear force bound	Describes correlation optimized linear force sensing. Dissipation fixes the residual force scale.	[5, 6]
26	Wineland spin squeezing criterion	$\xi_R^2 = N(\Delta J_\perp)^2/ \langle \mathbf{J} \rangle ^2 < 1$	Metrological witness	Includes contrast loss through the mean spin length. A value below unity supports sub projection noise Ramsey sensitivity.	[40, 26]
27	Quantum cooperativity	$C_q = C/\bar{n}_{\text{th}}, C = 4G^2/(\kappa\Gamma_m)$	Regime criterion	Uses a common high occupation convention. Values above unity indicate that quantum measurement dynamics compete with thermal decoherence.	[7]
28	Error corrected metrology condition	$\mathcal{S} = \text{span}_{\mathbb{C}}\{I, L_k, L_k^\dagger, L_k^\dagger L_j\}_{j,k}, G \notin \mathcal{S}$	Metrology architecture	The signal generator must lie outside the Lindblad operator span under Markovian noise. Clean ancillas and fast accurate control are assumed.	[48]
29	Squeezing after passive loss	$V_{\text{out}}/V_{\text{vac}} = \eta e^{-2r} + 1 - \eta$	Channel relation	Propagation and detection loss impose a floor on observable squeezing. Improve total efficiency before increasing source squeezing.	[52, 2]
30	Matched filter signal to noise ratio	$\text{SNR}^2 = 4 \int_0^\infty \tilde{s}(f) ^2 / S_n^{(1)}(f) df$	Estimator relation	Assumes a known waveform in stationary Gaussian noise. It converts a spectral budget into a task level metric.	Standard

Appendix B

Approach Selection Matrix

Approach	Best matched target	Primary control knobs	Gain mechanism	Typical failure modes	Verification
Complete input referred noise budget	Every platform	Transfer functions, reference planes, power spectral density conventions, covariance model	Identifies the dominant term and prevents invalid comparisons	Missing cross terms, unpropagated subtraction uncertainty, reference plane mismatch	Injection calibration, independent sweeps, residual analysis
Thermalization and filtering	Microwave, mechanical, spin, acoustic, and magnonic modes	Temperature, frequency, attenuation, anchoring, shielding	Reduces thermal occupation and environmental decoherence	Nonequilibrium photons, pump heating, weak anchoring	Mode thermometry, sideband asymmetry, temperature sweep
High quantum cooperativity	Cavity electromechanics and optomechanics	Coupling G , linewidths κ and Γ_m , occupation	Makes measurement dynamics compete with thermal decoherence	Heating, instability, nonlinear frequency shifts	Power sweep, damping measurement
Low preamplifier loss	Microwave and optical readout	Cable, circulator, coupler, escape efficiency, detector efficiency	Preserves encoded information before gain	Standing waves, reduced isolation, thermal emission	Cold insertion loss, mode occupation, gain calibration
Quantum-limited phase preserving gain	Unknown signal phase or both quadratures	Pump power, detuning, gain, impedance, bandwidth	Suppresses downstream electronics with the minimum half quantum added noise	Compression, pump leakage, gain ripple, reverse noise	Calibrated quantum noise measurement across gain and input power
Phase sensitive gain	Known signal quadrature	Pump phase, quadrature angle, gain	Avoids added noise in one selected quadrature	Phase drift, signal rotation, antisqueezing leakage	Quadrature sweep and covariance ellipse
Measurement strength optimization	Continuous position, force, and qubit readout	Probe power, cavity detuning, integration time	Balances imprecision against backaction or dephasing	Heating, saturation, nonlinear transitions	Full power sweep with component fit
Matched filtering and Bayesian estimation	Known waveform or informative prior	Noise model, waveform template, prior, latency	Extracts the available information efficiently	Model mismatch, estimator bias, prior domination	Injection recovery, interval coverage, independent validation set
Squeezed input	Optical, microwave, and interferometric phase sensing	Squeeze factor, angle, loss, phase lock	Reduces fluctuations in the signal quadrature	Loss, phase noise, antisqueezing	Detected squeezing, loss budget, task level gain
Frequency dependent squeezing	Broadband interferometry	Filter cavity detuning, linewidth, loss	Rotates the reduced noise quadrature across frequency	Filter loss, mode mismatch, lock error	Broadband noise ratio and angle reconstruction
Variational readout	Correlated continuous measurement	Homodyne angle, detuning, power	Uses the imprecision and backaction cross spectrum to cancel disturbance	Correlation loss, phase error, narrow cancellation band	Measured cross spectrum or covariance model
Backaction evasion	One oscillator quadrature	Two tone balance, phase, sideband resolution	Sends disturbance into the conjugate quadrature	Counterrotating leakage, imbalance, parametric instability	Quadrature resolved noise and backaction scaling
Quantum nondemolition measurement or speed meter	Repeated observable or low frequency force	Interaction symmetry, timing, optical topology	Couples to an observable preserved by the ideal dynamics	Parasitic noncommuting terms, delay mismatch, loss	Repeated measurement correlation and disturbance test
Coherent quantum noise cancellation	Broadband force sensing	Auxiliary resonance, damping, coupling, phase	Produces destructive interference between backaction responses	Susceptibility mismatch, auxiliary decoherence	Cancellation bandwidth and auxiliary noise measurement
Dynamical decoupling	Alternating current field sensing and coherence extension	Pulse timing, pulse fidelity, sequence order	Filters environmental noise outside the signal band	Pulse errors, reduced direct current sensitivity, heating	Filter function spectroscopy and control error sweep
Spin squeezing	Atomic clocks and ensemble sensors	Interaction strength, state orientation, contrast, readout	Reduces collective projection noise	Decoherence, local oscillator noise, readout noise	Wineland parameter and full clock stability
Distributed entanglement	Global network observables	Channel loss, synchronization, estimator weights, node coupling	Allocates shared quantum Fisher information to a global function	Unequal nodes, phase drift, local loss	Network information matrix and separable strategy comparison
Quantum error correction	Signal and noise with separable operator structure	Code, ancilla quality, syndrome rate, decoder latency	Removes correctable noise and preserves the signal generator	Signal suppression, gate error, slow correction cycles	Logical coherence and retained signal response
Measurement based feedback	State stabilization, cooling, and dynamic range	Filter, loop gain, delay, actuator noise	Conditions and controls the state using the measurement record	In loop squashing, instability, hidden actuation noise	Out of loop record and closed loop susceptibility
Reservoir engineering	State preparation and autonomous stabilization	Dissipative coupling, pump detuning, bath purity	Uses controlled dissipation to cool or stabilize a useful state	Pump heating, auxiliary loss, spectral overlap	State covariance, occupation, preparation time, pump sweep
Adaptive measurement	Wide prior, phase wrapping, limited dynamic range	Basis update, interrogation schedule, latency	Keeps the detector near an informative operating point	Model error, control quantization, uncounted adaptive resources	Posterior coverage, bias, resource normalized comparison

Appendix C

Terminological Roots, Conceptual Relations, Nomenclature, and Abbreviations

C.1 Terminology scaffold

A **linguistic root** records the historical word source, etymological lineage, or namesake from which a term derives. Each **conceptual root** identifies the prior idea that makes the term intelligible. **Related terms** name descendants, neighboring concepts, broader classes, narrower cases, or contrasts that clarify how each term sits within the surrounding knowledge system.

C.2 Selective etymology and interpretive value

Term	Origin or literal sense	Interpretive value in this handbook
Measurement	English <i>measure</i> descends through Anglo-French from Latin <i>mensura</i> and <i>metiri</i> , which refer to determining extent or quantity [54].	Measurement is treated as a complete comparison and inference process that connects a physical parameter, a reference, a record, and an estimator.
Metrology	French <i>métrologie</i> descends from Greek <i>metrologia</i> and <i>metron</i> , meaning measure [55].	The term denotes the science of measurement, including calibration, traceability, uncertainty, and comparison rules that extend beyond high precision.
Parameter	New Latin formed the word from <i>para-</i> and Greek <i>metron</i> , meaning measure [56].	A parameter characterizes the state, model, or process under study. In this handbook it is the physical quantity to be inferred and differs from a generic boundary.
Quantum	Latin <i>quantum</i> , the neuter form of <i>quantus</i> , asks “how much” [57].	The root recalls a quantity or amount. Modern quantum theory adds discrete excitations, noncommuting observables, superposition, entanglement, and channel constraints.
Transduction	Latin <i>transducere</i> means to lead across or transfer [58].	A transducer carries a signal or energy representation across physical domains. Transduction can occur without net gain, so it remains distinct from amplification.

Term	Origin or literal sense	Interpretive value in this handbook
Estimate and estimator	Latin <i>aestimare</i> means to value, appraise, or estimate [59].	An estimator is the rule that maps data to an inferred parameter. An estimate is the realized numerical output of that rule.
Spectrum	New Latin <i>spectrum</i> comes from Latin for an appearance or image [60].	A spectrum is an ordered representation of variance, amplitude, occupation, or response across frequency, energy, or another conjugate coordinate.
Stochastic	Greek <i>stochastikos</i> refers to skill in aiming or conjecturing from a target [61].	A stochastic model represents probability-governed dynamics. The process can retain structure, correlations, and a specific physical mechanism.
Phase	Greek <i>phasis</i> refers to an appearance, especially the appearance of a star or lunar phase [62].	In oscillatory systems, phase specifies position within a cycle relative to a declared reference. Relative phase carries physical information independently of whether absolute phase is physically defined.
Quadrature	The historical term concerns square construction and configurations separated by a right angle [63].	Field quadratures are orthogonal components separated by one quarter of a cycle. Their conjugate relation explains phase-sensitive readout and squeezing.
Coherence	The term follows Latin <i>cohaerentia</i> , associated with parts holding together [64].	Quantum coherence describes phase relations that remain mutually consistent and can generate interference. It differs from population retention and energy relaxation.
Dissipation	Latin <i>dissipare</i> means to scatter or disperse [65].	Dissipation transfers organized energy from selected degrees of freedom into environmental channels. Its response signature enters fluctuation-dissipation relations and ultimate force bounds.
Feedback	The modern engineering compound denotes returning part of a system output to its input or controller [66].	Feedback uses the measurement record to modify later dynamics. In-loop quieting remains distinct from improved out-of-loop information about an external signal.

C.3 Root-term families and conceptual relations

Family	Conceptual roots	Related or derived terms	Principal distinctions and cross-reference
Measurement and inference	Parameter, probe, observable, record, estimator	Accuracy, precision, sensitivity, resolution, uncertainty, bias, mean square error, calibration, traceability	Parameter versus estimate; accuracy versus precision; sensitivity versus resolution. See Chapter 1.
Probability and statistics	Random variable, probability density, expectation value, variance	Covariance, likelihood, prior, posterior, confidence interval, credible interval, Fisher information	Variance versus mean square error; likelihood as a function of the parameter versus probability over outcomes. See Chapters 1 and 6.
Linear response and transduction	Input, output, generalized force, coordinate, gain, susceptibility	Transfer function, bandwidth, reference plane, calibration tone, input referral, dynamic loading	Susceptibility describes physical response; a transfer function describes the complete input-output map. See Chapters 1 and 3.

Family	Conceptual roots	Related or derived terms	Principal distinctions and cross-reference
Noise and spectra	Fluctuation, correlation function, Fourier transform, stationarity	Power spectral density, amplitude spectral density, cross spectrum, thermal noise, technical noise, quantum noise, imprecision, backaction	One-sided versus two-sided spectra; symmetrized versus unsymmetrized spectra; added noise versus total measured noise. See Chapter 2.
Quantum state and dynamics	State, density operator, observable, commutator, Hamiltonian, channel	Quadrature, coherence, squeezing, entanglement, dephasing, dissipation, decoherence, measurement backaction	State uncertainty constrains preparation statistics; backaction describes detector-induced dynamics. See Chapters 1 and 5.
Metrological bounds	Resource, generator, estimator, error metric, benchmark	Shot noise limit, standard quantum limit, Heisenberg limit, quantum Fisher information, quantum Cramér–Rao bound, Holevo bound, quantum Ziv–Zakai bound	Architecture-dependent standards differ from optimized estimation bounds. Every comparison requires matched resources and assumptions. See Chapters 6 and 10.
Readout and amplification	Mode, loss, gain, bandwidth, added noise, efficiency	Phase-preserving amplification, phase-sensitive amplification, parametric gain, dispersive readout, measurement-induced dephasing, signal to noise ratio	Source noise differs from amplifier-added noise; pre-gain loss removes information that downstream gain cannot recover. See Chapter 8.
Control and stabilization	Measurement record, filter, controller, actuator, environment	Matched filter, Wiener filter, Kalman filter, Bayesian estimator, feedback, adaptive measurement, reservoir engineering, dynamical decoupling, error correction	In-loop noise squashing differs from external sensitivity improvement; control resources belong in the measurement contract. See Chapters 4 and 7.
Implementation interface	Materials loss, mode participation, quasiparticle occupation, nonlinearity, geometry, process variation	Cavity linewidth, dispersive shift, Kerr response, leakage, thermalization, electromagnetic parameter extraction, compression	Device physics supplies the calibrated parameters used by the common measurement framework. See the scope discussion in Chapter 1 and the readout treatment in Chapter 8.

C.4 Namesakes used as technical labels

Technical label	Namesake	Interpretive note
Fisher information	Ronald A. Fisher	The quantity measures local statistical distinguishability. Quantum Fisher information optimizes this local information over permitted measurements.
Cramér–Rao bound	Harald Cramér and C. R. Rao	The bound connects estimator variance to Fisher information under regularity and local unbiasedness assumptions.
Holevo Cramér–Rao bound	Alexander S. Holevo	The multiparameter bound accounts for incompatibility among optimal quantum measurements [31, 30].
Helstrom bound	Carl W. Helstrom	The bound gives the minimum binary quantum-state discrimination error for declared priors and states [32].

Technical label	Namesake	Interpretive note
Ziv–Zakai bound	Jacob Ziv and Moshe Zakai	The bound relates global mean square error to finite-separation discrimination and remains informative when local curvature is insufficient [28, 27].
Quantum Chernoff bound	Herman Chernoff	The quantum form governs the asymptotic error exponent for repeated binary state discrimination [33].
Wineland squeezing parameter	David J. Wineland	The parameter measures metrological spin squeezing and includes the loss of collective contrast [40].
Landauer principle	Rolf Landauer	The principle associates logically irreversible erasure with a minimum thermodynamic cost under its stated assumptions [39].
Heisenberg limit	Werner Heisenberg	In metrology, the label denotes an ideal resource scaling or bound whose precise form depends on the generator and resource convention. [24, 25].

C.5 Mathematical and physical symbols

Symbol	Name	Meaning in this handbook
θ	Parameter	Physical quantity inferred from the measurement record.
$\hat{\theta}$	Estimator output	Inferred value produced by a declared rule acting on the data.
$y(t)$	Measurement record	Classical detector output as a function of time.
ρ	Density operator	Quantum state used to predict outcome probabilities.
$\langle A \rangle$	Expectation value	Mean value of A over repeated trials or a quantum state.
$\text{Var}(A)$	Variance	Mean square spread of A around its expectation value.
$\text{Cov}(A, B)$	Covariance	Joint fluctuation measure for A and B .
$H_{y\theta}$	Transfer function	Complex gain and phase from the input parameter to the output.
χ_{xx}	Susceptibility	Linear response of coordinate x to a generalized force.
$S_{AB}[\omega]$	Unsymmetrized spectrum	Quantum spectrum whose operator order and frequency sign carry physical information.
$\bar{S}_{AB}[\omega]$	Symmetrized spectrum	Average of the two operator orderings used for variance and many detector noise budgets.
$S_{AA}^{(1)}(f)$	One sided spectrum	Engineering spectrum in hertz for positive frequencies.
$\bar{n}_{\text{th}}$	Thermal occupation	Bose–Einstein mean population of a mode.
N_{add}	Added noise quanta	Input referred noise contributed by an amplifier or readout chain.
η	Efficiency or transmission	Context identifies passive power transmission or information efficiency.
C	Cooperativity	Coherent interaction or measurement rate normalized to damping.
C_q	Quantum cooperativity	Cooperativity normalized to thermal occupation in the stated convention.
F_C	Classical Fisher information	Information available from a fixed measurement probability distribution.
F_Q	Quantum Fisher information	Maximum local Fisher information available from a quantum state.
ξ_R^2	Wineland squeezing parameter	Metrological spin squeezing measure that includes contrast loss.

Symbol	Name	Meaning in this handbook
Γ_φ	Dephasing rate	Rate at which phase coherence is lost during readout or environmental coupling.
κ	Cavity linewidth	Energy decay rate of an electromagnetic cavity mode.
Γ_m	Mechanical linewidth	Energy damping rate of a mechanical mode.

C.6 Abbreviations

Abbreviation	Expansion	Meaning in this handbook
RMS	Root mean square	Square root of the mean squared value.
MSE	Mean square error	Estimator error containing both variance and squared bias.
PSD	Power spectral density	Variance per unit frequency.
ASD	Amplitude spectral density	Square root of a power spectral density.
SNR	Signal to noise ratio	Dimensionless measure of signal separation relative to noise.
SQL	Standard quantum limit	Architecture dependent quantum benchmark defined by explicit assumptions.
SNL	Shot noise limit	Independent counting or independent probe benchmark.
HL	Heisenberg limit	Canonical ideal scaling proportional to $1/N$ under the stated generator normalization.
dB	Decibel	Logarithmic unit for power or amplitude ratios.
UQL	Ultimate quantum limit	Correlation optimized linear force bound set by dissipative response in the stated model.
QFI	Quantum Fisher information	Maximum local Fisher information available over permitted quantum measurements.
QFIM	Quantum Fisher information matrix	Multiparameter extension of quantum Fisher information.
QCRB	Quantum Cramér–Rao bound	Local estimator variance bound based on quantum Fisher information.
HCRB	Holevo Cramér–Rao bound	Weighted multiparameter precision bound that accounts for measurement incompatibility.
QZZB	Quantum Ziv–Zakai bound	Global Bayesian mean square error bound.
QCB	Quantum Chernoff bound	Asymptotic binary discrimination error exponent.
QND	Quantum nondemolition	Measurement designed to preserve the selected observable under repeated interrogation.
BAE	Backaction evading	Measurement that directs detector disturbance away from the selected observable.
CQNC	Coherent quantum noise cancellation	Auxiliary response engineered to cancel backaction coherently.
SLD	Symmetric logarithmic derivative	Operator used to construct quantum Fisher information.
GHZ	Greenberger–Horne–Zeilinger	Maximally correlated multipartite state used in metrology and information science.
HEMT	High electron mobility transistor	Low temperature semiconductor amplifier commonly used after a cryogenic first stage.

Abbreviation	Expansion	Meaning in this handbook
LIGO	Laser Interferometer Gravitational Wave Observatory	Large scale interferometric gravitational wave detector.
AC	Alternating current	Time varying signal with a nonzero oscillation frequency.
DC	Direct current	Constant or very low frequency signal component.

Appendix D

References and License

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